Reacting metals with acids

Acids are **corrosive** and eat some substances away. Some metals react when placed in acid. The piece of metal gets smaller, and bubbles of hydrogen gas are produced. The metal has not actually disappeared – it has changed into a new product, called a **salt**. As well as sodium chloride (common table salt), there are many other types of salts. The general **word equation** to show a reaction between an acid and a metal is:

metal + acid ⟶ salt + hydrogen gas

The first part of the name of the salt that forms comes from the metal and the second part comes from the acid used. For example, if calcium reacts with hydrochloric acid, the salt that is produced is called calcium chloride.

☞ **1** Cross out the incorrect words from the options provided.

a Sulfuric acid forms chloride/nitrate/sulfate salts.

b Nitric acid forms chloride/nitrate/sulfate salts.

c Hydrochloric acid forms chloride/nitrate/sulfate salts.

2 Read the passage below and then answer the questions that follow.

Michael was conducting an experiment. When he placed magnesium in hydrochloric acid (hydrogen chloride), bubbles appeared, and the magnesium got smaller and disappeared. When he placed a lit match above the test tube, there was a squeaky pop. Michael knew that this meant the gas was hydrogen.

a What were the two reactants in Michael's experiment?

_____

b Name one product.

_____

c Look at the word equation below. It shows that there are two products produced. What do you think the missing product is?

magnesium + hydrochloric acid (hydrogen chloride) ⟶ m_____ c_____ + hydrogen gas

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